# Exploiting Embedding Manifold of Autoencoders for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection is an important task in the remote sensing domain. Recently, researchers have shown great interest in deep learning-based methods because they can learn hierarchical, abstract, and high-level representations. However, the latent features learned from the autoencoder (AE) are not always able to reflect the intrinsic structure of hyperspectral data because the locality property is not considered during the learning process. In order to address this problem, a novel manifold constrained AE network (MC-AEN)-based hyperspectral anomaly detection method is proposed in this article. First, the manifold learning method is employed to learn the embedding manifold. Then, the latent representations are learned by an AE network with the learned embedding manifold constraints to preserve the intrinsic structure of hyperspectral data. Finally, the reconstruction errors are calculated to detect anomalies. The global reconstruction error from MC-AEN and the local reconstruction error from the learned latent representations are combined to fully utilize the learned knowledge for better detection performance. We test our proposed algorithm on three different real data sets. Experimental results on these three data sets show the superiority of our proposed method.**

***Index Terms*— Autoencoder (AE), global reconstruction, hyperspectral imagery (HSI), hyperspetral anomaly detection, local reconstruction, manifold learning.**

## I. Introduction

HYPERSPECTRAL imagery (HSI) records spectra that have fine wavelength resolution and covers a wide range of wavelengths from visible to near-infrared or short-wave infrared [1]. Hyperspectral imaging can measure contiguous spectral bands. Hence, each ground material of HSI has a continuous unique spectral curve, which makes it possible to identify the subtle differences of different ground objects [2], [3]. Due to this unique characteristic, HSI has been used in many fields [4]. Among them, hyperspectral anomaly detection is a very important and fundamental research topic.

The purpose of hyperspectral anomaly detection is to locate a target whose spectrum is significantly different from the surrounding background [5]. Objects in the hyperspectral image whose spectrum is significantly different from their surrounding pixels are regarded as hyperspectral anomalies. The whole image removes anomalies and the remaining objects can be seen as background [2], [6], [7]. Generally, anomalies are small objects with low probabilities of occurrence [8]. Hyperspectral anomaly detection can be generally modeled as a binary classification. It is an unsupervised technique compared with supervised target detection because it needs no prior information about targets or background [9], [10]. It can be also viewed as a special case of target detection. Since the spectral information of many ground materials cannot be known in advance, hyperspectral anomaly detection is more suitable for the practical situation and applied to many fields, such as civilian search and rescue [11], precision agriculture [12], and mineral exploration [13].

A major challenge in hyperspectral anomaly detection methods is how to construct an efficient representation for background [14]. The deep learning-based method can learn the hierarchical, abstract, and high-level representation, which is very helpful for target detection and recognition [15]. In the hyperspectral image processing domain, deep learning-based methods draw more and more attentions. They can be applied to various tasks, such as hyperspectral classification [16], change detection [17], hyperspectral anomaly detection [18], [19], and hyperspectral unmixing [20].

Autoencoder (AE) network is one of the most important unsupervised networks. Bati *et al*. [14] proposed a hyperspectral anomaly detection method based on AE, which first represented hyperspectral background with a different complexity by AE. Then, anomalies were detected by calculating reconstruction errors between the original image and the decoded image. Zhao *et al*. [18] proposed a stacked denoising AEs based hyperspectral anomaly detection method to automatically extract nonlinear deep features. Zhao and Zhang [21] proposed spectral–spatial stacked AEs based on low-rank and sparse matrix decomposition (LRaSMD) for hyperspectral anomaly detection. In order to alleviate the heavy training computational burden, Ma *et al*. [22] presented a deep AE-based maximum-distance-pixel-library algorithm for the online application. Deep belief networks (DBNs)

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are also applied to the hyperspectral anomaly detection [23]. Li *et al*. [19] proposed a CNN-based anomaly detection framework that is trained by the differences between pixel pairs. Anomalies are measured by calculating similarity scores of the output from CNN.

Although the above-mentioned deep learning-based methods have achieved promising results, an important issue has been ignored. The above-mentioned hyperspectral anomaly detection methods based on AEs or DBNs do not consider the local structural information of the hyperspectral data. The assumption of hyperspectral anomaly methods based on AEs or DBNs is that anomalies occur with a lower probability than background and contribute less to the training of the AE or DBN model [23]. Reconstruction errors of anomalies are larger than those of background. The learned latent representations from AEs or DBNs are global without considering the local geometry of the hyperspectral data. Hyperspectral anomaly detection methods regard pixels whose spectrum is significantly different from its neighbors as anomalies. According to this definition, the local characteristics can help improve detection accuracy. Fortunately, manifold learning is an effective approach to reduce nonlinear dimensionality, which can explore the intrinsic structure of data. In these cases, we propose a novel manifold constrained AE network (MC-AEN)-based hyperspectral anomaly detection method that explicitly incorporates the embedding manifold of hyperspectral data in the AE framework.

Since manifold learning is employed to constrain the training of AEs, the latent representation learned from AEs has the locality property that proves to be useful for the hyperspectral anomaly detection task. Hyperspectral anomaly detection methods based on AEs are global-based methods, which use the global reconstruction errors to detect anomalies. However, using only global reconstruction error cannot take advantage of the learned knowledge from MC-AEN. Therefore, the local-based hyperspectral anomaly detection method is applied to the learned latent representation from AEs, which also detects anomalies by calculating reconstruction errors. The final detection results are acquired by combining the global reconstruction error from the MC-AEN and the local reconstruction error from the learned latent representations.

The contributions of this article can be summarized as follows.

1) We propose an MC-AEN that is constrained by the embedding manifold learned from manifold learning to preserve the intrinsic structure of hyperspectral data.
2) We present a global–local-based hyperspectral anomaly detection method. The global reconstruction error calculated by the global-based method based on MC-AEN and the local reconstruction error computed by the local-based method on the learned latent representations are both considered to distinguish anomalies from the background more precisely.
3) We verify our proposed hyperspectral anomaly detection method on real data sets and obtain promising anomaly detection results.

This article is organized as follows. Section II will briefly discuss the main types of hyperspectral anomaly detection methods and manifold learning methods. In Section III, our proposed method will be described in detail. First, AEs are discussed. Second, MC-AENs are introduced. Finally, the global–local-based hyperspectral anomaly detection method is described in detail. In Section IV, the evaluation of the proposed method is verified on three real data sets. Section V is the conclusion.

## II. Background

### A. Hyperspectral Anomaly Detection

The main types of hyperspectral anomaly detection are briefly discussed in this section.

The well-known RX [24] algorithm calculates the Mahalanobis distance between the spectrum of the pixel under test and its neighbors to detect anomalies. The assumption of RX is that the background obeys the multivariate normal distribution. In order to better estimate the covariance matrix, many modified RX detectors have been proposed, such as kernel RX [25], weight RX [26], and others [27]. However, most of the data in the real world cannot meet the multivariate normal distribution.

Representation-based methods assume that a background pixel in the HSI can be represented by a linear combination of a prelearned dictionary or its neighbors while anomalies cannot [28], [29]. Sparse representation-based detectors [9], [30] and collaborative representation-based detectors [31], [32] are the two important types of representation-based methods. Reconstruction errors are calculated to distinguish anomalies.

LRaSMD-based hyperspectral anomaly detection methods assume that background of HSI data is low rank and anomalies are spatially sparse [33]. Anomalies are detected by computing the Euclidean distance on the sparse matrix (anomaly matrix). Many modified methods based on LRaSMD are proposed to improve detection accuracy. Zhang *et al*. [34] replace the Euclidean distance by the Mahalanobis distance. Xu *et al.* [35] decompose the background part of LRaSMD into the background dictionary and its corresponding coefficient matrix. Qu *et al*. [10] decompose the abundance vectors from spectral unmixing-based LRaSMD.

### B. Manifold Learning

Manifold learning methods are nonlinear dimensionality reduction methods [36]. They have the ability to discover the compact and nonlinear representation of the given high-dimensional data. Manifold learning methods can reveal the meaningful and intrinsic structure of the given high-dimensional data by preserving distances within local neighborhoods of the data [37], such as ISOMAP [38], locally linear embedding (LLE) [39], Laplacian eigenmaps [40], and diffusion maps [41]. LLE is an unsupervised method that aims to discover the low-dimensional, neighborhood-preserving embedding of the given high-dimensional data by exploiting the local symmetries of linear reconstructions.

The main steps of LLE are described as follows.

1) Compute the neighbors of each data point $\boldsymbol{X}_i$ in $\boldsymbol{X}$ space using K-nearest neighbors, where $X \in R^{D\times N}$, $X$ contains $N$ samples with D dimensionality.

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2) Compute the weights $\boldsymbol{W}_{ij}$ that best reconstruct each data point $\boldsymbol{X}_i$ from its neighbors, minimizing the following cost function:

$$\varepsilon(W) = \sum_i \left| X_i - \sum_j W_{ij} X_j \right|^2 \tag{1}$$

where $\boldsymbol{W}_{ij} = 0$ if $\boldsymbol{X}_j$ does not belong to the set of neighbors of $\boldsymbol{X}_i$.

3) Compute embedding coordinates $\boldsymbol{Y}_i$ best reconstructed by the weights $\boldsymbol{W}_{ij}$, minimizing the embedding cost function

$$\Phi(Y) = \sum_i \left| Y_i - \sum_j W_{ij} Y_j \right|^2 \tag{2}$$

where $Y \in R^{d \times N}$, $d \ll D$.

## III. Our Method

### A. AE

The AE network is a typical unsupervised deep neural network, and its output is equal to the input. The aim of AEs is to learn a latent-space representation (encoding) of the input. An AE network is usually composed of an encoder and a decoder. The encoder can be modeled by the encoder function $h = f(x)$ that compresses the input $\boldsymbol{X}$ into a latent-space representation $\boldsymbol{H}$. Generally, the dimension of the latent representation is smaller than that of the input. The decoder is modeled by the decoder function $r = g(h)$ that aims to reconstruct the input $\boldsymbol{X}$ from the latent-space representation $\boldsymbol{H}$. Hence, the AE can be described by the function $r = g(f(x))$ as a whole, where $r$ is required to as close to the original input $x$. Generally, the AE is trained by minimizing the reconstruction error loss $L = \|r - x\|_2$. AEs are trained to preserve as much information as possible when input is run through the encoder and then the decoder but are also trained to make the new representation have various nice properties.

### B. Manifold Constrained AE

Pixels whose spectrum is significantly different from their neighbors are regarded as anomalies for hyperspectral images. According to this definition, hyperspectral anomaly detection is very concerned with the structural information of hyperspectral data. However, the input of the original AEs for hyperspectral anomaly detection is the spectral information of pixels and does not take the structural information into account.

Since manifold learning methods can preserve distances within local neighborhoods of the data [37], they can reveal the meaningful and intrinsic structure of the given high-dimensional data. Hence, manifold learning methods can provide structural information about the input data. In the learned manifold, the samples with similar spectral signatures are clustered and those with different spectrums are far apart. Inspired by this idea, we try to use the manifold learning method to constrain the learning of the latent representation of AEs. The aim is to make the latent representation with local structural information that is helpful for the hyperspectral anomaly task.

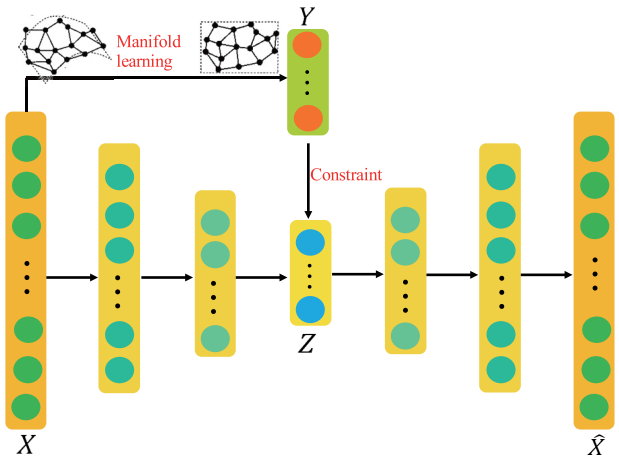



Fig. 1. Flowchart of our proposed MC-AEN.

Fig. 1 shows our proposed MC-AEN. The proposed network contains of two parts: the encoder and the decoder. The architecture of MC-AEN is as follows: FC[80]-FC[40]-FC[embedding size]-FC[40]-FC[80], where FC[$n$] denotes a fully connected layer with $n$ outputs and the embedding size represents the dimension of the latent representation, which is decided by the manifold learning method.

From Fig. 1, it can be seen that the latent representation extracted from the AE is constrained by the embedding manifold learned from the manifold learning method. LLE is employed as a manifold learning method in this article. The reason why we use a manifold learning method to constrain the latent representation learning of the AE is that the latent representation learned from the AE does not contain the structural information. Fortunately, manifold learning method can provide a new way to get the structural information or reflect the intrinsic structure of the input data. Hence, we employ the learned embedding representation $\boldsymbol{Y}$ from LLE to constrain the latent representation $\boldsymbol{Z}$ learning from the AE to make the latent representation $\boldsymbol{Z}$ with the structural information to reflect the intrinsic structure of the input $\boldsymbol{X}$.

The loss function of classic AE is defined as follows:

$$L = \frac{1}{2N} \sum_{i=1}^{N} \|\boldsymbol{x}_i - \hat{\boldsymbol{x}}_i\|^2 + \frac{\lambda}{2} \sum_{l=1}^{L-1} \|\boldsymbol{W}^{(l)}\|_F^2 \tag{3}$$

where the first term denotes the mean square error and the second term is a regularization term. $N$ represents the number of samples. $\boldsymbol{x}_i$ and $\hat{\boldsymbol{x}}_i$ are the $i$th saw input sample and its corresponding reconstructed sample, respectively. $L$ is the number of layers in the AE. $\boldsymbol{W}^{(l)}$ represents the weights between layer $l$ and layer $l+1$. $\lambda$ is the weight decay parameter.

In order to constrain the latent representation $\boldsymbol{Z}$ learning by the learned embedding representation $\boldsymbol{Y}$, we have modified the above-mentioned classical loss function, which is defined as

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follows:

$$L=\frac{1}{2N}\sum_{i=1}^{N}\|\boldsymbol{x}_i-\hat{\boldsymbol{x}}_i\|^2+\frac{1}{2N}\sum_{i=1}^{N}\|\boldsymbol{y}_i-\boldsymbol{z}_i\|^2+\frac{\lambda}{2}\sum_{l=1}^{L-1}\|\boldsymbol{W}^{(l)}\|_F^2 \tag{4}$$

where $\boldsymbol{y}_i$ and $\boldsymbol{z}_i$ denote the embedding representation and the latent representation of the $i$th sample $\boldsymbol{x}_i$. Since $\boldsymbol{Y}$ has the structural information, $\boldsymbol{Z}$ can contain the structural information by letting $\boldsymbol{Z}$ fit $\boldsymbol{Y}$. MC-AEN is proposed to obtain the latent representation that can preserve the intrinsic structure of hyperspectral data.

The proposed MC-AEN is described in detail as shown in Algorithm 1.

**Algorithm 1** MC-AEN

**Input:** Training set $\boldsymbol{X}$, the embedding manifold $\boldsymbol{Y}$, the number of layers for manifold constrained AE network $L$

**Output:** the reconstruction sample set $\hat{\boldsymbol{X}}$

1: Pre-train every hidden layer of manifold constrained AE network.

2: Initialize the weights $\theta=\{\boldsymbol{W}^{(l)},\boldsymbol{b}^{(l)}\}_l$ of the manifold AE by the pre-training result.

3: Optimize the parameters $\{\boldsymbol{W},\boldsymbol{b}\}$ of the network by fine-tuning cost Eq. 4 with back-propagation.

4: Compute the reconstruction sample set $\hat{\boldsymbol{X}}$,
$\hat{\boldsymbol{X}}=f(\boldsymbol{W}^{(L-1)}X+\boldsymbol{b}^{(L-1)})$.

From Algorithm 1, our proposed MC-AEN needs the pre-training procedure. Hinton *et al*. [42] first develop a greedy layerwise pretraining procedure for the restricted Boltzmann machines. This can improve the optimization of the neural network. In MC-AEN, each layer is trained in an unsupervised manner whose input is the output of the previous layer. The obtained parameters $\theta=\{\boldsymbol{W}^{(l)},\boldsymbol{b}^{(l)}\}_l$ can be performed as the initialized parameters of MC-AEN. Compared with the random weights initialization, the weights parameters acquired from the pretraining procedure can improve results.

In this article, we pretrain every hidden layer as a denoising AE [43], [44]. The loss function is defined as follows:

$$L=\frac{1}{2N}\sum_{i=1}^{N}\|\boldsymbol{x}_i-\hat{\boldsymbol{x}}_i\|^2+\frac{\lambda}{2}\sum_{l=1}^{L-1}\|\boldsymbol{W}^{(l)}\|_F^2+\beta\sum_{j=1}^{s}KL(\rho\|\hat{\rho}_j) \tag{5}$$

where the first, second, and third terms are the mean square error, a regularization term, and the sparsity loss, respectively. $N$ represents the number of samples. $\boldsymbol{x}_i$ and $\hat{\boldsymbol{x}}_i$ are the $i$th saw input sample and its corresponding reconstructed sample, respectively. $L$ is the number of layers in the AE. $\boldsymbol{W}^{(l)}$ represents the weights between layer $l$ and layer $l+1$. $\lambda$ and $\beta$ are the weight decay parameters. $s$ indicates the number of hidden units. $\rho$ is a sparsity parameter, usually a small value close to zero, while $\hat{\rho}_j$ indicates the average activation of hidden unit $j$. $KL(\rho\|\hat{\rho}_j)$ denotes the Kullback–Leibler divergence between $\rho$ and $\hat{\rho}_j$. This constraint imposes that each hidden unit responds only to specific input patterns in the data.

### C. Global–Local-Based Hyperspectral Anomaly Detection

In general, anomalies are sparsely distributed in the whole hyperspectral image, and anomalies are small objects with low probabilities of occurrence [45]. Hence, the vast majority of the input to the AE are background pixels. In this case, AE-based hyperspectral anomaly detection methods essentially learn the pattern of background pixels by training the network. When anomalies are used as input, the learned network that represents the background pattern cannot model anomalies preciously. That is because the pattern of anomalies deviates from that of background pixels. The global reconstruction error $E_{\text{global}}$ of the $i$th pixel is calculated to detect anomalies, which is defined as

$$E^i_{\text{global}}=\|\boldsymbol{x}_i-\hat{\boldsymbol{x}}_i\|^2,\quad i=1,\ldots,N \tag{6}$$

where $\boldsymbol{x}_i$ is the original sample. $\hat{\boldsymbol{x}}_i$ is the reconstructed sample. $N$ is the number of pixels in the whole hyperspectral image.

According to the definition of hyperspectral anomaly detection, the spectrum of the anomaly is significantly different from those of their neighbors. The definition of hyperspectral anomaly detection is highly related to the local region. Hence, the local property is very useful for the hyperspectral anomaly detection task. The latent representation learned from AEs constrained by manifold learning has the locality property. In the learned latent representation space, similar samples are closer and different samples are farther than those in the original spectral space. In the learned latent representation space, the reconstructed representation $\hat{\boldsymbol{z}}_i$ of the $i$th pixel can be represented by the following formula:

$$\hat{\boldsymbol{z}}_i=\sum_{j\neq i}^{k}\boldsymbol{W}_{ij}\boldsymbol{z}_j,\quad i=1,\ldots,N \tag{7}$$

where $\boldsymbol{W}$ are the reconstruction weights and are solved by manifold learning method LLE. $k$ is the number of neighbors. If a pixel $\boldsymbol{z}_i$ in the learned latent representation space is a background pixel, the reconstruction error between $\boldsymbol{z}_i$ and $\hat{\boldsymbol{z}}_i$ computed by (8) should be small. While if this pixel is an anomaly, it cannot be represented by the neighbors that are background pixels accurately. Hence, the local reconstruction error of the $i$th pixel calculated in the embedding feature space can be utilized to detect anomalies, which is defined as

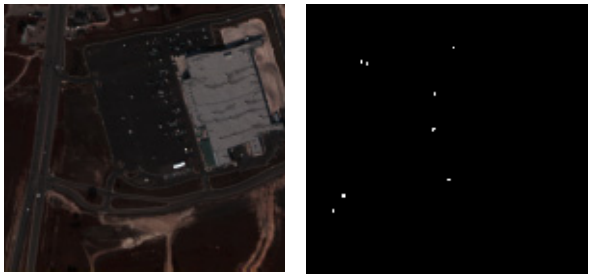
$$E^i_{\text{local}}=\|\boldsymbol{z}_i-\hat{\boldsymbol{z}}_i\|^2. \tag{8}$$

The proposed method in this article detects anomalies by calculating reconstruction errors. The learned knowledge cannot be fully utilized with only one global reconstruction error or local reconstruction error. In order to guarantee the good detection performance, we combine the global reconstruction error from manifold learning constrained AE network and the local reconstruction error from the latent representations to jointly decide whether this pixel belongs to the anomaly. Hence, the overall reconstruction error $E$ of the $i$th pixel includes $E^i_{\text{global}}$ and $E^i_{\text{local}}$, which is defined as

$$E^i=(1-\alpha)E^i_{\text{global}}+\alpha E^i_{\text{local}} \tag{9}$$

where $\alpha$ is the tradeoff parameter, $0\leq\alpha\leq 1$. The detailed analysis of this tradeoff parameter $\alpha$ is shown in

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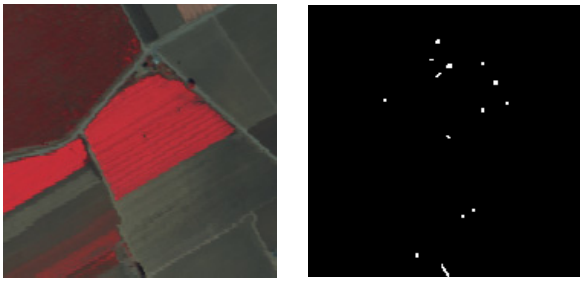
(a) (b)

Fig. 2. (a) Pseudocolor image of HYDICE urban data. (b) Ground truth.



(a) (b)

Fig. 3. (a) Pseudocolor image of AVIRIS airplane data. (b) Ground truth.



(a) (b)

Fig. 4. (a) Pseudocolor image of Salinas data. (b) Ground truth.

Section IV-D2. If the reconstruction error is larger than the predefined threshold $\delta$, it will be detected as the anomaly, and otherwise, it belongs to the background.

## IV. Experiments

### A. Data Sets

The effectiveness of our proposed MC-AEN-based hyperspectral anomaly detection is verified on three different hyperspectral images. These three hyperspectral data sets will be briefly introduced as follows.

1) *HYDICE Urban Data Set*: The hyperspectral digital imagery collection experiment (HYDICE) urban data set is one of the most widely used HYDICE image data sets, which is collected by the HYDICE sensor. This data set is available on the website http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetsArticleView/tabid/9254/Article/476681/hypercube. The size of the original hyperspectral image is $307 \times 307$. It has 210 bands with the spectral wavelengths ranging from 400 to 2500 nm. The following bands (1–4, 76, 87, 101–112, 136–153, and 197–210) are removed because of the water absorption regions, low SNR, and bad bands. The remaining 162 bands are used to preserve more useful information. A subimage with a size of $150 \times 150$ in the upper right of the original image is cropped in our experiment in this article. The types of anomalies in this data set are usually cars and roofs. The visualization of the HYDICE urban data and its corresponding ground truth is illustrated in Fig. 2, which can be found in this article [2].
2) *AVIRIS Airplane Data Set*: The Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) airplane data set is acquired by the AVIRIS from San Diego, CA, USA. The AVIRIS airplane data set has 20 m spatial resolution and 10-nm spectral resolution with the spectral wavelengths ranging from 370 to 2510 nm. The size of the AVIRIS airplane with 224 spectral bands is $100 \times 100$ in the spatial dimension. The following bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224) are removed by considering the water absorption regions, low SNR, and bad bands. The remaining 189 bands are employed to retain the information. There are three airplanes in the image, which are viewed as anomalies. The visualization of the AVIRIS airplane data set and its corresponding ground truth is illustrated in Fig. 3, which can be obtained in this article [3].
3) *Salinas Scene Data Set*: The Salinas data set is also collected by AVIRIS and is in Salinas Valley, CA, USA. The data set is available on the website http://www.ehu.es/ccwintco/. The Salinas data set was recorded in 1998 with 224 bands. The size of the original image is $512 \times 217$. A subimage with a size of $180 \times 180$ is cropped from the original image in this article to test the effectiveness of our proposed method. The Salinas scene includes different types of targets, such as bare soils, vegetables, and vineyard fields. The following bands (108–112, 154–167, and 224) are removed by taking low SNR and water absorption into account. Fig. 4 shows the visualization of the Salinas data set and its corresponding ground truth, which can be obtained in this article [46].

### B. Competitors

The proposed MC-AEN-based anomaly detector is compared with the state-of-the-art methods as follows:

1) Global RX (GRX) [24] is a classical hyperspectral anomaly detection method. The GRX method is based on the assumption that the background obeys a multivariate normal distribution. The GRX method detects anomalies by calculating the Mahalanobis distance between the pixel under test and its neighbors. GRX estimates global background statistics.
2) Local RX (LRX) [24] estimates the local background statistics. The other aspects of LRX are consistent with those of GRX.

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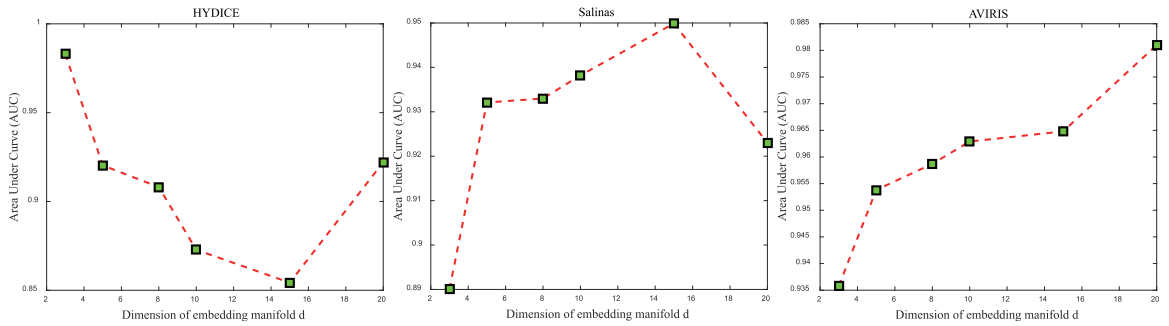



Fig. 5. AUC values with the change of the dimension of embedding manifold $d$.

3) Background of joint sparse representation (BJSR)-based method [30] just uses the background bases to construct the overcompleted background dictionary. BJSR is employed to estimate the adaptive orthogonal background complementary subspace. Then, the reconstruction errors based on the background dictionary are computed to detect anomalies.
4) LRaSMD [34] acquires the background through the LRaSMD method. Then, the Mahalanobis distance based on the background statistics is calculated to detect anomalies.
5) Modified collaborative representation detector (MCRD) [32] gets more representative neighbors by removing pixels whose spectrum is significantly different from a majority of pixels in the background. The background is estimated by the remaining pixels. Finally, anomalies are detected by the residual error.
6) AE-based method [14] represents the hyperspectral background of a different complexity with the AE. Then, anomalies are determined by calculating reconstruction errors between the original image and the decoded image.

### C. Evaluation Criteria

Receiver operating characteristic (ROC) curve is one of the most important performance assessment tools for target detection applications [47]. Hence, ROC curve is also used to verify the effectiveness of our proposed MC-AEN-based hyperspectral anomaly detection method. The ROC curve is plotted by setting different thresholds of the true positive rate (TPR) and false positive rate (FPR). FPR refers to the probability of false alarm and is the $X$-axis. TPR means the probability of detection and is the $Y$-axis. The formula of TPR and FPR can be defined as follows:

$$\begin{aligned} \mathrm{TPR} &= \frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FN})} \\ \mathrm{FPR} &= \frac{\mathrm{FP}}{(\mathrm{FP}+\mathrm{TN})} \end{aligned} \tag{10}$$

where TP means true positive, TN refers to true negative, FP is false positive, and FN represents false negative.

ROC curve just shows the performance of our proposed method from the qualitative perspective. In order to further verify the effectiveness of our proposed method from the quantitative perspective, AUC value is calculated, which means the area under ROC curve (encircled by the ROC curve and the false alarm rate axis). AUC value providing a numerical accuracy measure is used to analyze the detection performance quantitatively. When the AUC value is 1, it stands for the perfect test.

### D. Parameter Settings

*1) Dimension of Embedding Manifold $d$:* The dimension of embedding manifold $d$ is a very important parameter for the manifold learning method. In our proposed MC-AEN-based hyperspetral anomaly detection method, the dimension of the embedding manifold $Y$ and the dimension of the latent representation $Z$ are both $d$. The value of $d$ is tuned manually in the experiment. When learning the embedding manifold, the number of neighbors $k$ of LLE is set to 48, and the global reconstruction error is calculated to detect anomalies. The value range of this parameter $d$ is set to [3, 5, 8, 10, 15, 20]. The AUC values with the change of the parameter $d$ on HYDICE, AVIRIS, and Salinas data sets are illustrated in Fig. 5. It can be obviously seen that the optimal dimensions of the embedding manifold are different for different data sets. This is determined by the intrinsic structure of the hyperspectral data. It can be observed from Fig. 5 that the optimal dimensions of the embedding manifold $d$ are 3, 15, and 20 for HYDICE urban, AVIRIS airplane, and Salinas data sets, respectively.

*2) Tradeoff Parameter $\alpha$:* The tradeoff parameter $\alpha$ can be used to balance the global reconstruction error and the local reconstruction error. The value of $\alpha$ can reflect which term has a greater contribution to the overall reconstruction error. The value of $\alpha$ is tuned manually in our experiment. These two parameters, the number of neighbors $k$ used in LLE and the dimension of the embedding manifold $d$, are very important parameters that should be set in advance. The values of $k$ and $d$ are 48 and 3 for the HYDICE urban data set. For the AVIRIS airplane data set, they are 32 and 15. for the Salinas data set, the value of $k$ and $d$ are 32 and 20, respectively. The range of the tradeoff parameter $\alpha$ we set in the experiment is [0, 0.01, 0.05, 0.1, 0.5, 1].

The parametric experiment is done on the three different data sets to verify the effect of the tradeoff parameter $\alpha$. The quantitative detection results, AUC values obtained from the

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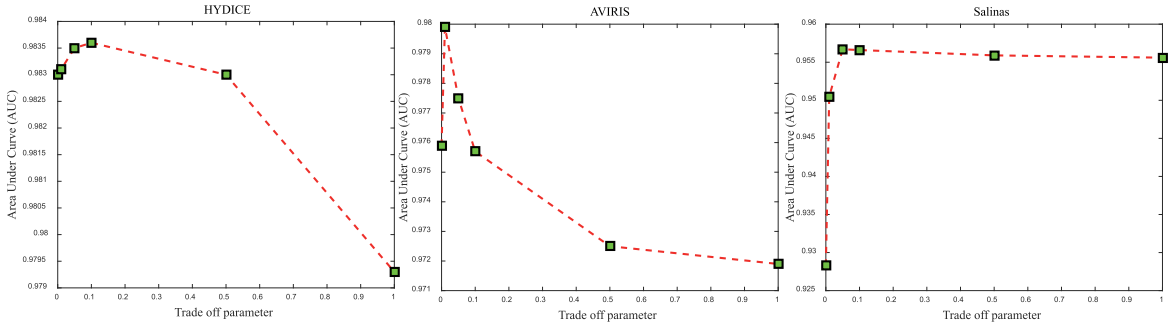



Fig. 6. AUC values with the change of the tradeoff parameter $\alpha$.

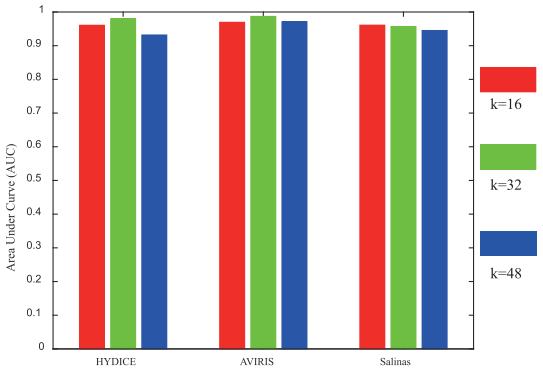



Fig. 7. AUC values with the change of the number of neighbors $k$.

overall reconstruction error, with the change of $\alpha$ are shown in Fig. 6. It can be seen that the effect of the tradeoff parameter $\alpha$ on AUC values for three different data sets is relatively regular. However, the optimal values of $\alpha$ are different for the three data sets. They are 0.1, 0.01, and 0.1 for HYDICE, AVIRIS, and Salinas, respectively.

*3) Number of Neighbors k:* The number of neighbors $k$ is a very important parameter in the LLE method, which affects the accuracy of learning the low embedding manifold. Hence, it is necessary to verify the effect of the parameter $k$. The parameter of the dimension of embedding manifold $d$ should be set in advance. It is 15, 15, and 20 for HYDICE, AVIRIS, and Salinas, respectively. The range of this parameter $k$ is set to [16, 32, 48]. Fig. 7 shows the quantitative results, AUC values acquired from the overall reconstruction error, with the change of the number of neighbors $k$.

It can be seen that the number of neighbors $k$ indeed has an effect on the detection accuracy. Moreover, its impact on three different data sets is different. The optimal values of $k$ are 32, 32, and 16 for HYDICE, AVIRIS, and Salinas, respectively.

### *E. Detection Performance*

In this section, we first compare our proposed method with the state-of-the-art methods listed in Section IV-B to prove the effectiveness of both qualitative and quantitative perspectives. Then, we will discuss the results of three data sets in detail.

The performances of LRX, BJSR, and MCRD listed in Section IV-B are related to the window size. In our experiment, the window size of LRX, BJSR, and MCRD are $W_{\text{out}} = 15 \times 15$ and $W_{\text{in}} = 3 \times 3$ for HYDICE urban, AVIRIS airplane, and Salinas data sets.

*1) Results of HYDICE Urban Data:* The visualized display of detection results on the HYDICE urban data set is shown in Fig. 9. We can observe from Fig. 9 that the number of anomalies detected by BJSR and LRaSMD is very small. Some anomalies are detected as the background pixels by mistake; therefore, BJSR and LRaSMD have a high miss detection rate. BJSR focuses on building a pure background dictionary. The size of anomalies in AVIRIS airplane and Salinas data sets is larger than those of the HYDICE data set. Hence, it is easier to build a pure background dictionary for the AVIRIS airplane and Salinas data sets. Therefore, the performance of BJSR is the poorest on HYDICE dataset. Compared with BJSR and LRaSMD, GRX, AE-based method, and MCRD can detect more anomalies and reduce the miss detection rate to a certain extent. However, they still miss some anomalies. Our proposed method and LRX can get better performance than other methods. Moreover, our proposed method gets a much better result than LRX because it almost detects all anomalies and the discrimination between anomalies and background pixels is relatively high.

Fig. 9 just visualizes the performance of our proposed method. In order to do the accurate estimation, we plot ROC curves for the HYDICE urban data set to make a further qualitative comparison, as shown in Fig. 8(a). It can be clearly seen that the ROC curve of our proposed method does not always stay above the other curves. When FPR is larger than $10^{-2}$, it stays over the other curves. Since ROC curves have inersections, it is necessary to further compute the AUC values to verify the performance of our proposed method.

The AUC values on HYDICE urban data set are listed in the second column of Table I. The local-based method, such as LRX and MCRD, gets better performance than the global-based methods, such as GRX, BJSR, and LRaSMD. Although our proposed MC-AEN-based method with the global reconstruction error belongs to the global-based method, the performance of our proposed method with the global reconstruction error is better than the other local-based method because of considering the intrinsic structure learned by manifold learning method. Our proposed global–local-based hyperspectral anomaly detection method achieves the best detection result by considering global and local reconstruction errors simultaneously.

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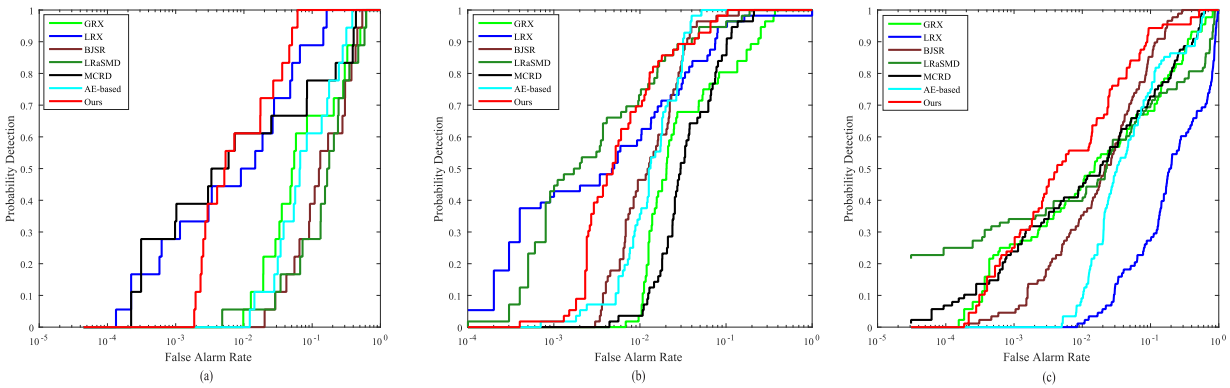



Fig. 8. ROC curves of hyperspectral anomaly detection algorithm for three different data sets. (a) HYDICE urban data. (b) AVIRIS airplane data. (c) Salinas data.

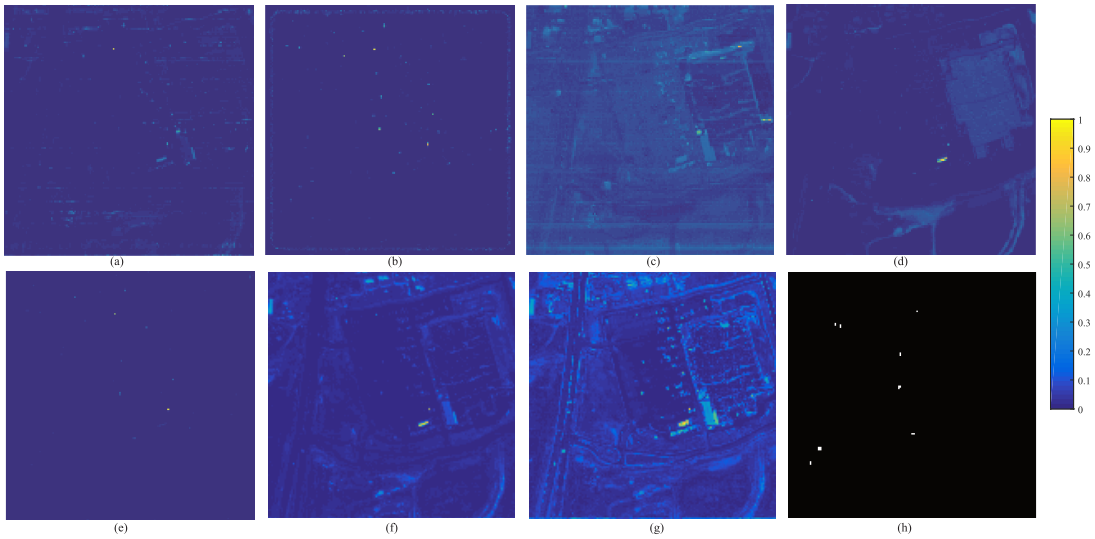



Fig. 9. Visualization of the detection results on the HYDICE urban data set. (a) GRX. (b) LRX. (c) BJSR. (d) LRaSMD. (e) MCRD. (f) AE-based method. (g) Ours. (h) Ground truth.

TABLE I

AUC VALUES OVER STATE-OF-THE-ART METHODS AND OUR PROPOSED ANOMALY DETECTION METHOD FOR THREE DIFFERENT DATA SETS

| | Data Sets | | |
|---|---|---|---|
| Methods | HYDICE | AVIRIS | Salinas |
| GRX | 0.8500 | 0.9357 | 0.8872 |
| LRX | 0.9678 | 0.9651 | 0.6357 |
| BJSR | 0.7988 | 0.9810 | 0.9533 |
| LRaSMD | 0.7790 | 0.9854 | 0.8187 |
| MCRD | 0.9071 | 0.9512 | 0.8972 |
| AE-based | 0.8871 | 0.9829 | 0.8929 |
| Global Loss | 0.9830 | 0.9759 | 0.9570 |
| Local Loss | 0.9079 | 0.9710 | 0.9323 |
| Ours | **0.9836** | **0.9871** | **0.9608** |

*2) Results of AVIRIS Airplane Data:* Fig. 10 shows the visualization of detection results on the AVIRIS airplane data set. The performance of GRX is worst for its highest miss detection rate. Anomalies of AVIRIS airplane data set only contain airplanes, while anomalies of the Salinas data set contain bare soils, vegetables, and vineyard fields. Thus, the anomaly types of the Salinas data set are more complex than that of the AVIRIS airplane data set. LRX has a higher performance on the AVIRIS airplane data set than the Salinas data set. LRX and MCRD detect more anomalies than GRX, but the detected anomalies have a low anomalous intensity. BJSR, LRaSMD, and AE-based methods achieve better detection results than LRX and MCRD. However, BJSR, LRaSMD, and AE-based method detect many background pixels as anomalies by mistakes; therefore, they have a relatively high false detection rate. Our proposed global–local-based method performs best in all methods. The anomalies detected by our proposed global–local-based method have a high anomalous intensity, and the number of background pixels that are detected as anomalies has decreased.

ROC curves are plotted as shown in Fig. 8(b) to further verify the performance of our proposed method from the quantitative perspective. The ROC curve of our proposed method stays over curves of some methods, such as GRX, MCRD, BJSR, and AE-based method. It can be seen from Fig. 8(b) that there are many crossovers among all methods, and it is hard to judge which algorithm performs well.

AUC values are calculated to further judge the performance. The third column of Table I shows AUC values on the AVIRIS

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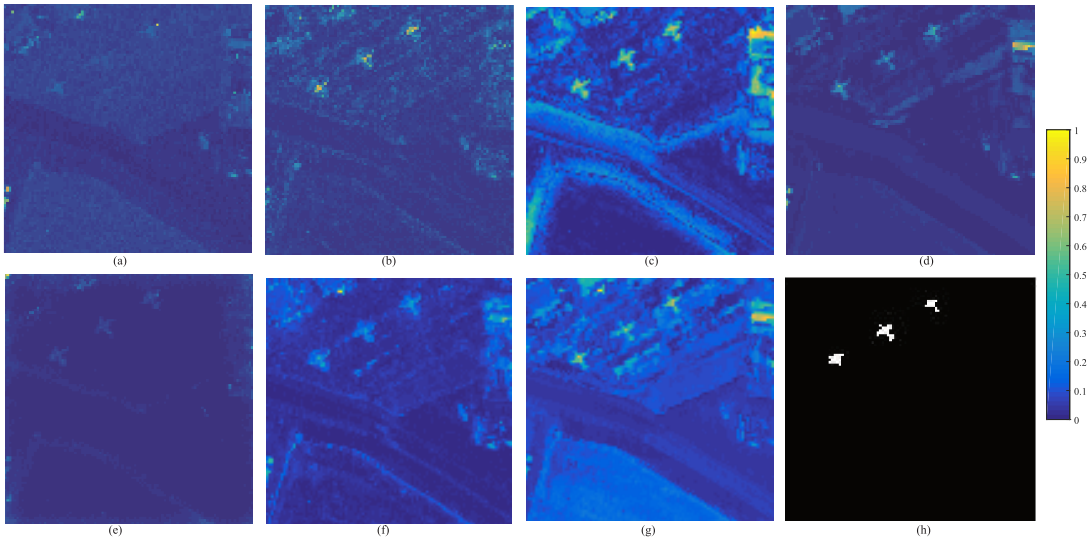



Fig. 10. Visualization of the detection results on the AVIRIS airplane data set. (a) GRX. (b) LRX. (c) BJSR. (d) LRaSMD. (e) MCRD. (f) AE-based method. (g) Ours. (h) Ground truth.

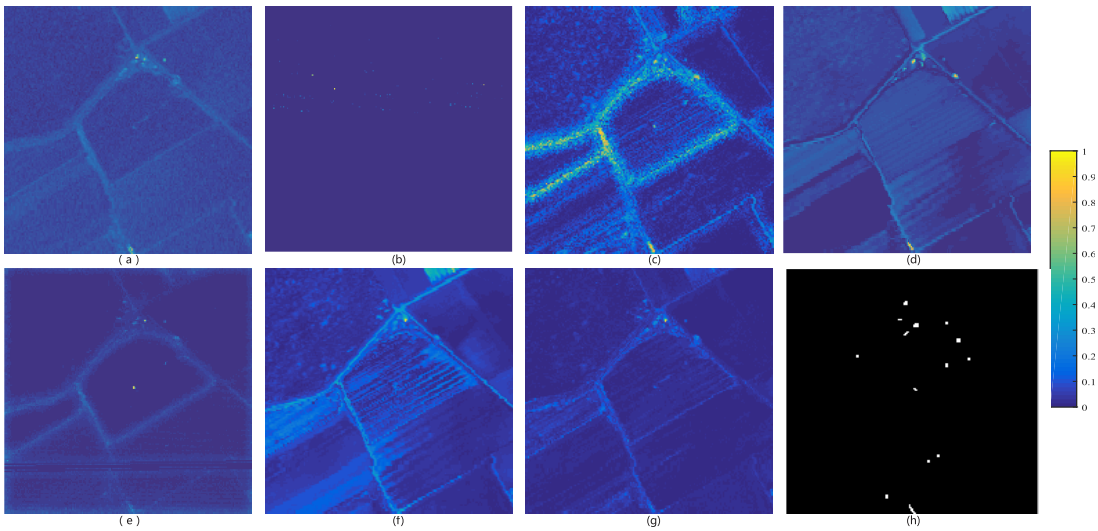



Fig. 11. Visualization of the detection results on the Salinas data set. (a) GRX. (b) LRX. (c) BJSR. (d) LRaSMD. (e) MCRD. (f) AE-based method. (g) Ours. (h) Ground truth.

airplane data set. It can be obviously seen that all methods acquire good detection results and AUC values except GRX are larger than 0.95. Our proposed global–local-based method gets the highest AUC value, 0.9871. AUC values of the MC-AEN-based hyperspectral anomaly detection method with global reconstruction error and local reconstruction error are 0.9759 and 0.9710, respectively. It proves that the combination of the global reconstruction error and the local reconstruction error is effective for the anomaly detection task. AUC values of GRX, LRX, BJSR, AE-based method, and MCRD are 0.9357, 0.9678, 0.9810, 0.9854, 0.9829, and 0.9512, respectively. The quantitative experimental results, AUC values, are consistent with the conclusion obtained from the visualization of detection results and ROC curves.

*3) Results of Salinas Data:* The detection results of the Salinas data set is visualized in Fig. 11. LRX misses most of the anomalies, while LRaSMD falsely detects many background pixels as anomalies. The performance of the GRX, BJSR, MCRD, and AE-based method is better than that of LRX and LRaSMD. Our proposed global–local-based method acquires the best performance.

Although Fig. 11 has already showed the effectiveness of our proposed global–local-based method directly, it illustrates the detection results from the qualitative perspective. The quantitative analysis should also be considered to make an accurate estimation. ROC curves for the Salinas data set are plotted in Fig. 8(c). It can be obviously seen that the ROC curve of our proposed method stays over the other curves.

The ROC curves of AE-based method, MCRD, LRaSMD, BJSR, and GRX have intersections. Hence, AUC values are calculated to further verify the effectiveness. The fourth column of Table I shows AUC values on the Salinas data set. LRX acquires the worst performance, and the AUC value of LRX is only 0.6357 and is consistent with the ROC curve. The

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size of anomalies in the Salinas data set is larger than those of the HYDICE data set. The performance of the LRX algorithm is related to the size of the double concentric window. The window size of LRX is $W_{\text{out}} = 15 \times 15$ and $W_{\text{in}} = 3 \times 3$ for the Salinas data sets in our experiment. Hence, the performance of LRX is much poorer on the Salinas data set than the HYDICE data set.

AUC values of GRX, LRaSMD, MCRD, and AE-based method are 0.8842, 0.8187, 0.8972, and 0.8929, respectively. BJSR performs better than the GRX, LRaSMD, MCRD, and AE-based method. AUC values of our proposed MC-AEN-based hyperspectral anomaly detection method with the global reconstruction error and the local error are 0.9570 and 0.9323, which are larger than most of the other comparison methods. It is obvious that our proposed global–local-based method achieves the best performance and the AUC value is 0.9608. From the above-mentioned analysis, the conclusions obtained from the visualization of detection results and ROC curves are consistent with AUC values. This proves our proposed global–local-based hyperspectral anomaly detection method is superior to other methods.

Although our proposed MC-AEN-based hyperspectral anomaly detection method can achieve better performance compared with other methods, it still has some limitations. On one hand, it requires a time-consuming and complex pretraining process. On the other hand, it still cannot correctly classify anomalies hardest to detect. Anomalies whose spectrum is very similar to those of background can be regarded as anomalies hardest to detect. In the original feature, the distance between anomalies hardest to detect and background is too small to detect these pixels as anomalies correctly. In the learned latent representation space, although the distance between anomalies hardest to detect and background has increased, it is still not large enough to detect these pixels as anomalies correctly.

## V. Conclusion

In this article, we propose a novel hyperspectral anomaly detection method-based MC-AEN. The manifold learning method is employed to constrain the learning of latent representations, which aims to make the latent representations with locality characteristic. The locality property is very useful for the hyperspectral anomaly detection task based on the definition of hyperspectral anomaly detection. In order to take advantage of the learned knowledge, we combine the global reconstruction error from MC-AEN and the local reconstruction error from the learned latent representations to guarantee better detection performance. The validity of our proposed algorithm is verified on three different real data sets. The extensive experimental results show that our proposed method can acquire better performance compared with other methods. In future work, we plan to learn the embedding manifold and the latent representations simultaneously.

## References


[1] W. Wang, B. Zhao, F. Feng, J. Nan, and C. Li, "Hierarchical subpixel anomaly detection framework for hyperspectral imagery," *Sensors*, vol. 18, no. 11, p. 3662, 2018.

[2] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 2, pp. 3123–3134, Dec. 2015.

[3] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Dec. 2018.

[4] D. Manolakis and G. S. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[5] S. Khazai, A. Safari, B. Mojaradi, and S. Homayouni, "An approach for subpixel anomaly detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 769–778, Apr. 2013.

[6] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[7] S. Li, K. Zhang, Q. Hao, P. Duan, and X. Kang, "Hyperspectral anomaly detection with multiscale attribute and edge-preserving filters," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1605–1609, Oct. 2018.

[8] Y. Niu and B. Wang, "Hyperspectral anomaly detection based on low-rank representation and learned dictionary," *Remote Sens.*, vol. 8, no. 4, p. 289, 2016.

[9] D. Ma, Y. Yuan, and Q. Wang, "A sparse dictionary learning method for hyperspectral anomaly detection with capped norm," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 648–651.

[10] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[11] M. T. Eismann, A. D. Stocker, and N. M. Nasrabadi, "Automated hyperspectral cueing for civilian search and rescue," *Proc. IEEE*, vol. 97, no. 6, pp. 1031–1055, Jun. 2009.

[12] D. Haboudane, J. R. Miller, E. Pattey, P. J. Zarco-Tejada, and I. B. Strachan, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, 2004.

[13] F. A. Kruse, J. W. Boardman, and J. F. Huntington, "Comparison of airborne hyperspectral data and EO-1 Hyperion for mineral mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1388–1400, Jun. 2003.

[14] E. Bati, A. Çalışkan, A. Koz, and A. A. Alatan, "Hyperspectral anomaly detection method based on auto-encoder," *Proc. SPIE*, vol. 9643, Oct. 2015, Art. no. 96430N.

[15] J. J. DiCarlo, D. Zoccolan, and N. C. Rust, "How does the brain solve visual object recognition?" *Neuron*, vol. 73, no. 3, pp. 415–434, 2012.

[16] A. Romero, C. Gatta, and G. Camps-Valls, "Unsupervised deep feature extraction for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1349–1362, Mar. 2016.

[17] J. Liu, M. Gong, K. Qin, and P. Zhang, "A deep convolutional coupling network for change detection based on heterogeneous optical and radar images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 545–559, Mar. 2018.

[18] C. Zhao, X. Li, and H. Zhu, "Hyperspectral anomaly detection based on stacked denoising autoencoders," *Proc. SPIE*, vol. 11, no. 4, 2017, Art. no. 042605.

[19] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[20] Y. Su, A. Marinoni, J. Li, J. Plaza, and P. Gamba, "Stacked nonnegative sparse autoencoders for robust hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1427–1431, Sep. 2018.

[21] C. Zhao and L. Zhang, "Spectral-spatial stacked autoencoders based on low-rank and sparse matrix decomposition for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 92, pp. 166–176, Aug. 2018.

[22] N. Ma, Y. Peng, S. Wang, and D. Liu, "Hyperspectral image anomaly targets detection with online deep learning," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf. (I2MTC)*, May 2018, pp. 1–6.

[23] N. Ma, Y. Peng, S. Wang, and P. H. W. Leong, "An unsupervised deep hyperspectral anomaly detector," *Sensors*, vol. 18, no. 3, p. 693, Feb. 2018.

[24] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

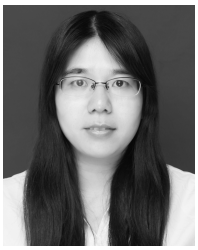
[25] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.


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[26] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.
[27] B. Du and L. Zhang, "Random-selection-based anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1578–1589, May 2011.
[28] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.
[29] Y. Yuan and Q. Wang, "Hyperspectral anomaly detection via iscriminative feature learning with multiple-dictionary sparse representation," *Remote Sens.*, vol. 10, no. 5, pp. 745-1–745-21, 2018.
[30] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.
[31] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.
[32] M. Vafadar and H. Ghassemian, "Anomaly detection of hyperspectral imagery using modified collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 577–581, Apr. 2018.
[33] W. Sun, C. Liu, J. Li, Y. M. Lai, and L. Weiyue, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *Proc. SPIE*, vol. 8, no. 1, 2014, Art. no. 083641.
[34] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.
[35] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.
[36] P. Yan, W. Zhang, B. Turkbey, P. L. Choyke, and X. Li, "Global structure constrained local shape prior estimation for medical image segmentation," *Comput. Vis. Image Understand.*, vol. 117, no. 9, pp. 1017–1026, 2013.
[37] G. Mishne, U. Shaham, A. Cloninger, and I. Cohen, "Diffusion nets," *Appl. Comput. Harmon. Anal.*, vol. 47, no. 2, pp. 259–285, 2017.
[38] J. B. Tenenbaum, V. de Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *Science*, vol. 290, no. 5500, pp. 2319–2323, Dec. 2000.
[39] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.
[40] M. Belkin and P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Comput.*, vol. 15, no. 6, pp. 1373–1396, 2003.
[41] R. R. Coifman and S. Lafon, "Diffusion maps," *Appl. Comput. Harmon. Anal.*, vol. 21, no. 1, pp. 5–30, Jul. 2006.
[42] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, 2006.
[43] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn.*, 2008, pp. 1096–1103.
[44] Q. V. Le, J. Ngiam, A. Coates, A. Lahiri, B. Prochnow, and A. Y. Ng, "On optimization methods for deep learning," in *Proc. 28th Int. Conf. Int. Conf. Mach. Learn.*, 2011, pp. 265–272.
[45] Y. Niu and B. Wang, "A novel hyperspectral anomaly detector based on low-rank representation and learned dictionary," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2016, pp. 5860–5863.
[46] Y. Yuan, Q. Wang, and G. Zhu, "Fast hyperspectral anomaly detection via high-order 2-D crossing filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 620–630, Feb. 2015.
[47] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems: An overview of practical algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.



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